Apache OpenOffice

Version 4.1

Draw Guide

AOO Documentation Team

Chapter 7

Working with 3D Objects

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Note for Mac Users

Some keystrokes and menu items differ on a Mac from those used in Windows and Linux. The table below gives some common substitutions for the instructions in this chapter. For a more detailed list, see the application Help.

Windows/Linux	***Mac equivalent***	***Effect***
Tools > Options menu selection	**OpenOffice > Preferences**	Access setup options
Right-click	*Control+click*	Open context menu
Ctrl (Control)	⌘ (Command)	Used with other keys
F5	*Shift+⌘+F5*	Open the Navigator
F11	*⌘+T*	Open Styles & Formatting window

Contents

Creating 3D Objects

Although Draw does not match the functionality of leading drawing or picture editing programs, it is capable of producing and editing very good 3D drawings and pictures.

Draw offers two different types of 3D objects: *3D bodies* and *3D shapes*. Depending on the type you choose, there are different options for further editing the object (rotation, illumination, perspective). 3D shapes are simpler to set up and edit than 3D bodies, but 3D bodies allow for more customization.

Creating 3D Bodies

You can produce 3D bodies in three ways: extrusion, body rotation, and using ready-made objects. In this chapter, these methods are called Variation 1, 2, and 3. Two further variations, extruding Basic Shapes and converting text to 3D, are also described.

Variation 1: Extrusion

First, draw one of the common Draw objects, for example, a square/rectangle, circle/ellipse, or a text box using the  or  icons on the Drawing toolbar or Sidebar. Next, choose **Modify > Convert > To 3D** (or right-click on the object and choose **Convert > To 3D**, or click the  icon) to produce a 3D object from the 2D surface. Figure 1 shows two examples.

The  icon is not usually visible on the Drawing toolbar. To make it visible, select **Visible Buttons** from the toolbar menu at the right end, then click the icon.

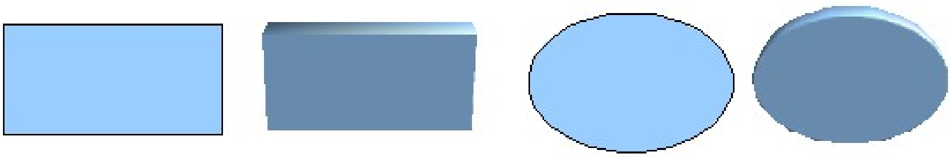


Figure 1: Extruding 3D objects from 2D surfaces

The process of moving parallel surfaces to create a 3D object is known as *extrusion*. In this case, the 2D surface is moved forward, “out of” the drawing level. At the same time, the object is slightly tilted, and central projection is turned on so that one can better recognize it. Draw uses a default extrusion value (the body depth) based on the 2D object's size. The value can be changed after the extrusion (see “Editing 3D Objects” on page 8).

Variation 2: Body Rotation

Choose a common drawing object, for example, a non-black line. Then change this into a rotation body. Draw provides two methods to do this.

Method 1. Click the icon  in the Drawing toolbar (this icon also usually needs to be made visible) or choose **Modify > Convert > To 3D Rotation object**. With this rotation method, the axis of rotation coincides with the left edge of the enclosing selection rectangle, through the green rectangle handles (Figure 2).

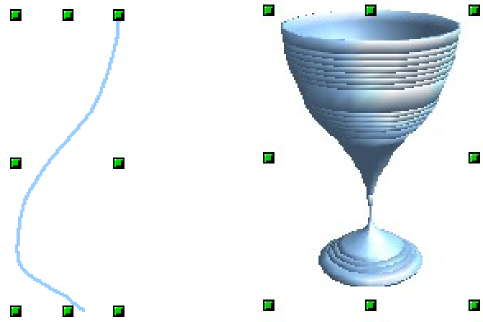


Figure 2: Rotation body created using variation 2.1

Method 2. Click the icon  on the *Effects* pull-down menu on the Drawing toolbar. This icon can also be accessed from the Mode menu, under **View > Toolbars > Mode**. Notice that this icon lacks the curved red arrow of the fixed-axis rotation icon (Figure 3).

With this method, you can change the location of the rotation axis, which appears as a dotted line with two white circular endpoints.

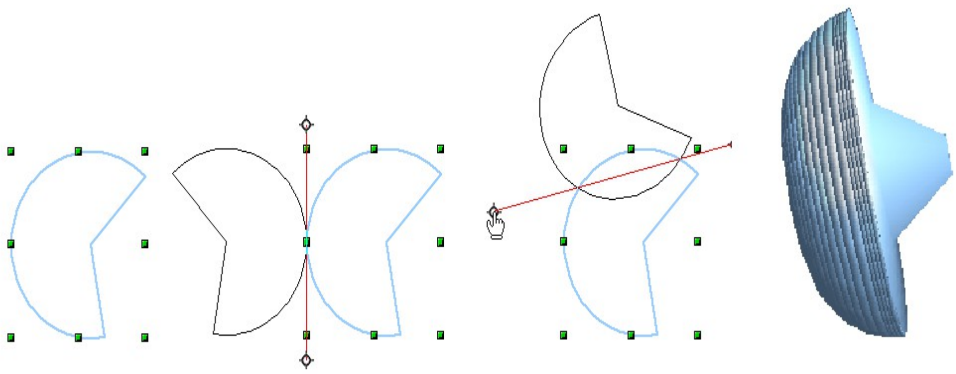


Figure 3: Rotation body created using variation 2.2

Click a white endpoint and drag it to move the axis to the desired position. (You may need to move both ends to achieve this.) The outline shows how the figure will be rotated. When you click the figure again, the rotation is applied, and a new 3D body is produced.

Variation 3: Using Ready-made Objects

Use the 3D Objects toolbar/pull-down menu (Figure 4). To activate this toolbar, click **View > Toolbars > 3D Objects**.

Figure 4: The 3D Objects toolbar

If you add the icon  to the Drawing toolbar, the 3D Object bar will be available as a dropdown menu or a floating toolbar.

After choosing the type of object, left-click on the starting point, and drag the mouse diagonally until the outline of the object is the size you want.

After releasing the mouse button, the 3D object appears. To change its height: width ratio, keep the *Shift* key pressed while dragging the mouse. Most 3D objects are constructed by rotation, but cubes and spheres are special types of 3D objects that are directly defined in the program code.

3D Scenes

Variations 1–3 all produce a *3D scene*. If you click a 3D scene, the status bar text shows *3D scene selected*. Such a scene is actually a group of objects.

If you constructed the scene using one of the methods above, it consists of the 3D body as a single element. 3D scenes can, however, be extended to include other 3D objects, as described in “Combining Objects in 3D Scenes” on page 26.

As mentioned in Chapter 6 (Editing Pictures), you can access individual elements of the group using **Modify > Enter group** or the context menu. The status bar text then changes and shows the type of each individual element selected, for example, *Sphere selected* or *Extrusion object selected*.

Producing 3D Shapes

Since the release of Version 2.0 of OpenOffice.org in 2005, Draw has contained a type of drawing object known as a (Basic) Shape. A special method of extrusion exists for these shapes.

Variation 4: Extrusion of Basic Shapes

Figure 5: The Basic Shapes toolbar

You use the Basic Shapes toolbar shown in Figure 5 (or another shape toolbar) to produce 2D surfaces. The Basics Shapes toolbar is available as a tear-off toolbar on the Sidebar or the Drawing toolbar. Shapes such as cylinders or cubes are technically possible but not very useful, because they produce curiously curved images. If you have drawn a shape, the last icon  on the Drawing toolbar is active. A click on this icon can transform a 2D surface into a 3D object (see Figure 6).

Extrusion of a shape does not create a new object type; it simply changes the shape’s appearance. All the object’s properties and settings are retained. You can use this button to toggle between a 2D and a 3D view. The object's properties and settings are not lost when switching between views.

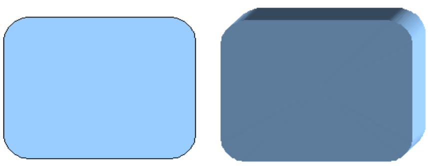


Figure 6: 3D objects from basic shapes

If you click such an object, the 3D-Settings toolbar (Figure 7) appears. If this does not open automatically, enable it in **View > Toolbars > 3D-Settings**.

Figure 7: 3D-Settings toolbar

The first icon  corresponds to the icon on the Drawing toolbar. This icon also works as a toggle switch. After switching to 2D, the 3D-Settings toolbar is hidden again. To change the object back to 3D, you must use the icon  on the Drawing toolbar.

Variation 5: Fontwork

For text, you can also use shapes from the Fontwork Gallery. These produce extrusion objects similar to those from Variation 4.

To open the Fontwork Gallery shown in Figure 8, use the icon  on the Drawing toolbar.

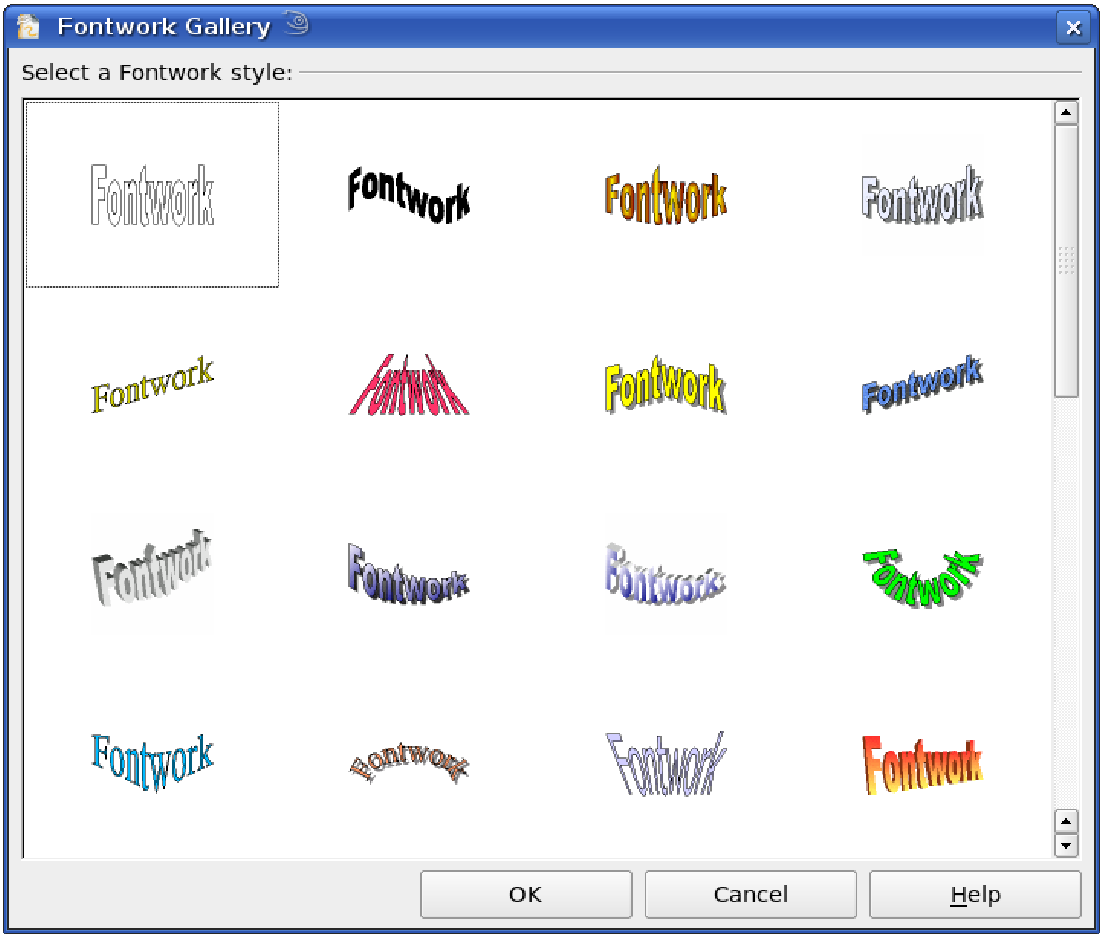


Figure 8: Extrusion Shapes from the Fontwork Gallery

Editing 3D Objects

Rotating 3D Objects

Procedure for 3D bodies

The Rotation command used for 2D objects also works with 3D objects, but due to the additional axis, there are some differences in editing 3D objects (see Figure 9). The procedure for selecting the object is identical to that used for a 2D object.

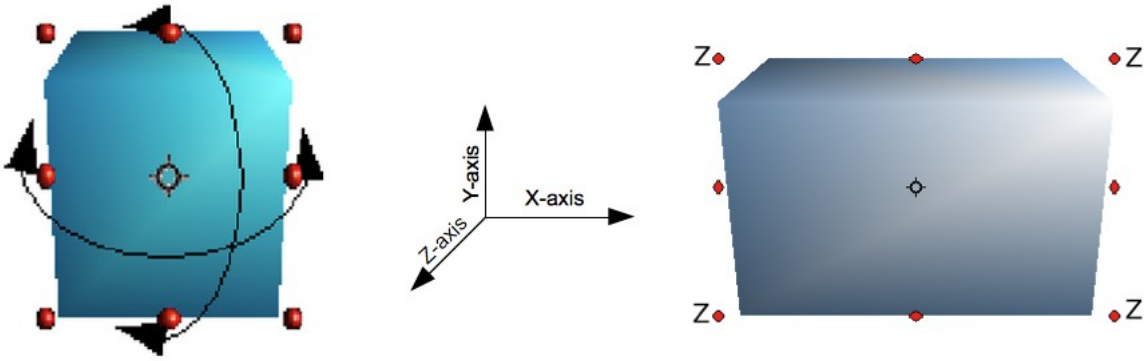


Figure 9: 3D object rotation

You can rotate the object about each axis (X, Y, Z). The X and Y axes are those parallel to the edges of the drawing layer, while the Z axis comes out of the page. It is not possible to change the axis orientation.

The three axes are not shown directly, but the symbol indicates their intersection point.

If you want to …	You must …
… rotate the object about the X or Y axis.	… put the mouse cursor over the object. With the left mouse key pressed, you can rotate the object as you wish. Moving one of the red points at the <u>middle</u> of an edge allows you to rotate the object about only one axis. Note that the cursor initially has the shape for a (2D) shearing movement, but pressing the mouse button changes it to a rotation cursor.
… rotate the object about the Z axis.	… move the handle at one <u>corner</u> point while the left mouse button is pressed. Rotation about the Z axis is independent of the rotation angle setting in the Position and Size dialog
… move the axis intersection point.	… simply drag the Symbol to the desired location. The point is located by default in the middle of the object.

These rotations can be applied to the entire 3D scene or to individual objects within it.

Procedure for 3D Shapes

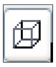
Objects produced using variations 4 and 5 (see above) can only be rotated about the Z axis when using the three methods described in the previous section. This rotation is applied to the underlying 2D object. It is also possible to rotate the 3D object in the same way as a 2D object, using **Format > Position and Size > Rotation** (shortcut key for Position and Size is *F4*), and specify the pivot point and the desired degree of rotation.

Shape objects have their own procedure for rotation about the X and Y axes. If you have activated the 3D-Settings toolbar (under **View > Toolbars**), it becomes active when you select the 3D object. Icons 2 to 5 on this toolbar (see Figure 10) rotate the object in 5-degree increments about the X and Y axes.

Figure 10: 3D-Settings toolbar with rotation icons indicated

3D settings for 3D Bodies

3D Effects Dialog – General Buttons

The 3D Effects dialog offers a wide range of possible settings for 3D objects produced with variations 1 to 3 (see previous sections). To open the dialog, right-click on the object and choose **3D Effects** from the pop-up menu. You can also activate a 3D Effects icon  on the Standard toolbar or add it to another toolbar using **Customize Toolbar > Add > Category Options > 3D Effects**.

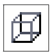
Tip	If you cannot see **3D Effects** in the pop-up menu, you may have a 3D object extruded from a basic shape. Try clicking the Extrusion On/Off icon in the Drawing Toolbar.

The dialog can be docked in a similar manner to the Navigator or Template windows.

The possible settings are arranged in various thematic categories on separate pages, accessible via the buttons at the top of the window, which function similarly to tabs. To apply the settings you have altered, click the Assign button, which is the checkmark icon at the right end of Figure 11. This applies *all* the changes you have made to the object on the other pages of the dialog, as well as the current one.

Figure 11: Upper part of the 3D Effects dialog

Note	Only the selected effects are assigned to the object in question. There is no object conversion; thus, a cylinder cannot be transformed into a ring through the application of a 3D effect. However, it is possible to change the appearance to resemble a wooden or metal body. By the assignment of a 3D effect, 2D objects are transformed into 3D objects.

In order for the Effects dialog to take over all the current properties of the object, you must click the  button. If you deactivate this button before you leave an object and then click it again when you open the Effects dialog for another object, the settings carry over from the first object to the second. This can be used to transfer favorite settings from one object to another, as all the settings are brought over to the new object. In normal use, however, the icon should remain active.

At the bottom-left of the dialog is another row of buttons (see Figure 12), available on each page.

Figure 12: Buttons for geometric transformations

The first two buttons correspond to the menu commands **Modify > Convert > To 3D / To 3D Rotation Object**. When the Effects dialog for a 3D object is called up, these buttons are inactive. Selecting a 2D object activates it. The third icon switches a perspective view of the object on or off.

 Converts the selected 2D object to an extrusion body.

 Converts the selected 2D object to a rotation body.

 Switches between a central projection and a perspective projection.

In a *central* projection, parallel edges appear to meet at a common point in the distance, as shown on the button symbol. *Parallel* projection preserves all parallel edges; a procedure often used to produce oblique figures. The switching process applies to the entire 3D scene.

With central projection (see Figure 13), Draw creates the object with three vanishing points. The parameters for central projection are set (indirectly) through the camera settings on the *Shading* dialog page.

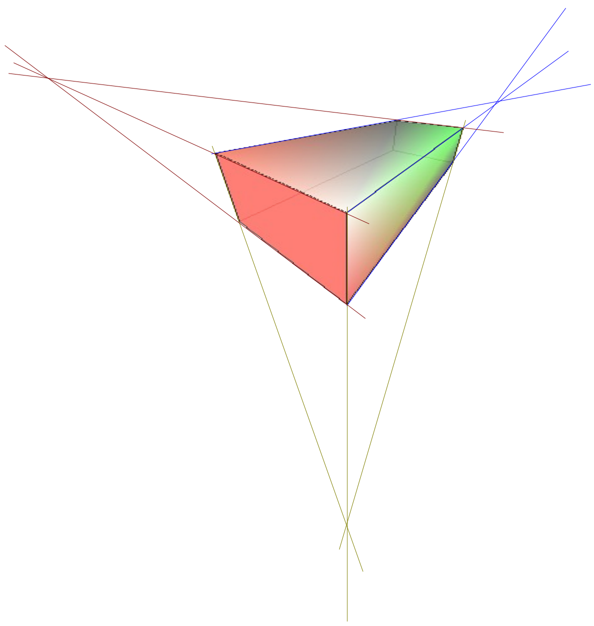


Figure 13: Figure shown using central projection. For clarity, the projection lines have been added.

3D Effects - Geometry

On the Geometry page, you can make changes to the geometry of a 3D object. This page is opened with the Geometry button  in the upper part of the 3D Effects dialog (Figure 14).

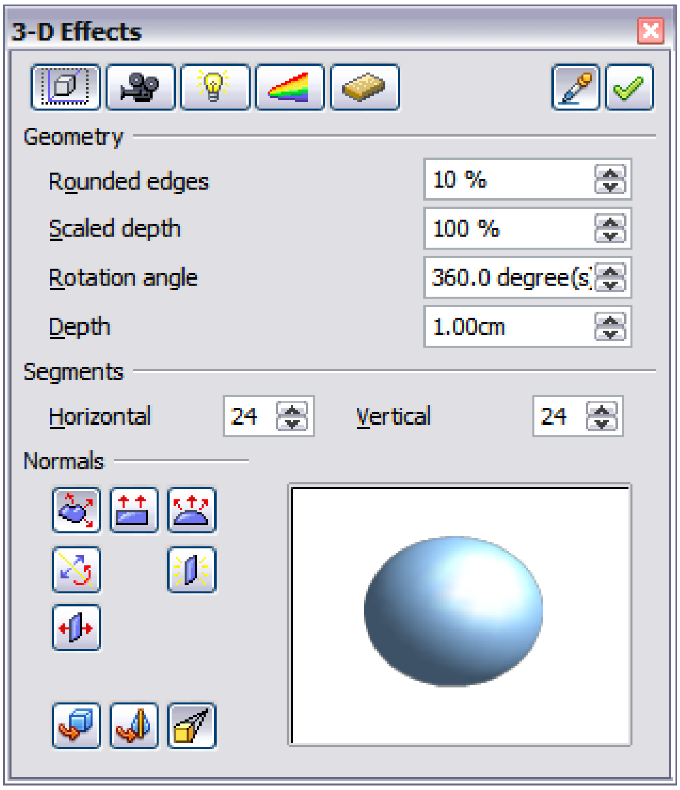


Figure 14: Geometry page

In the first example, the body's depth (length) is to be changed. This is possible only if you created it through extrusion. To illustrate: draw a circle and convert it according to Variation 1 into a 3D object (a flat cylinder); see examples a and b in Figure 15.

If necessary, select the cylinder, open the 3D Effects dialog, make sure the Geometry page is active, change the parameter Depth to 3.5cm, and click on the **Assign** icon. The cylinder should now resemble object c in Figure 15.

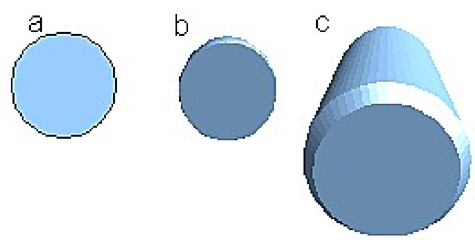


Figure 15: Conversion by extrusion of a 2D object (a) into a 3D object (b) and then changing its depth (c)

This parameter has no effect on a rotation body or on ready-made 3D objects, although the input field remains active.

With the *Rounding* parameter, you can specify how strongly the edges of the 3D object are rounded. Select (if necessary) the lengthened cylinder again, and use the

3D Effects dialog to change the rounding to 30%. The cylinder should now resemble that in Figure 16.

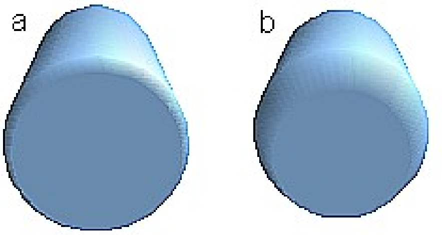


Figure 16: Edge rounding of 10% (a) and 30% (b)

This parameter also has interesting effects when you convert text into a 3D object.

The *Scaled Depth* parameter sets the front-to-back size ratio for a 3D object produced by extrusion. The front side of such an object always protrudes from the original surface; the rear side is the original surface of the 2D object (i.e., the 2D exit surface), even if the object has been rotated in the meantime.

By default, scaling is set to 100%, so both surfaces scale by the same amount. If you set this scaling to 50%, the cylinder becomes a truncated cone (see Figure 17).

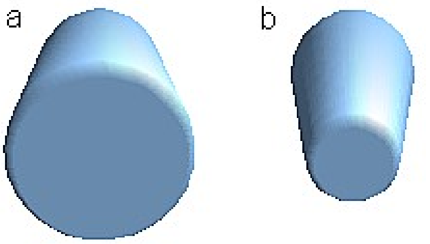


Figure 17: Cylinder with 50% scaling

The diameter of the front side is 50% that of the rear side. It is also possible to create the reverse effect, with the rear side smaller than the front, by using a scaling depth greater than 100%.

For rotation bodies, this parameter influences the width of the surface parallel to the axis of rotation. At the end of the rotation, the figure's surface width is given by the scaling depth. The distance to the rotation axis remains unchanged. In Figure 18, a line is rotated to create a 3D object with a scaling depth figure of 0%.

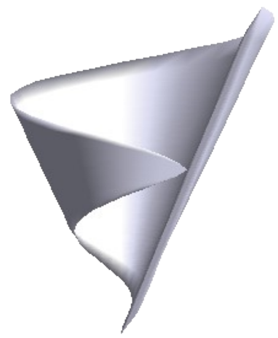


Figure 18: Scaling depth of a rotation body

The *Rotation Angle* parameter is only available for rotation bodies. With this parameter, you can create a segment of a complete rotation body by choosing an angle less than 360 degrees. Figure 19 shows a hemisphere with a rotation angle of 270°.

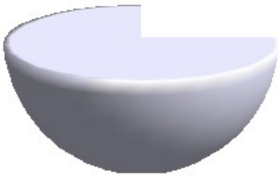


Figure 19: Hemisphere with a rotation angle of 270°

The *Horizontal* and *Vertical* Segments parameters define the number of segments out of which Draw builds spheres and rotation objects[1]. For rotation objects, the horizontal segments are more important. The vertical segments influence the degree of edge rounding.

In Figure 20, the left sphere is made up of 10 horizontal and vertical segments, while the right sphere has 25 segments. More segments yield a smoother surface, but such objects take longer to render the figure on screen. By default, spheres and hemispheres are constructed with 24 segments. For a square pyramid, you need four horizontal segments.

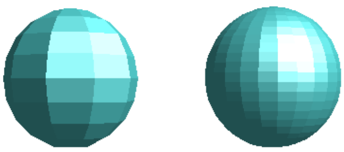


Figure 20: Sphere made from 10 segments (left) and 25 segments (right)

These are properties belonging to individual objects. If you apply the segment settings to a 3D scene, all objects in the scene are modified accordingly.

If you extrude an unfilled circle or intersecting lines with a filling, the result may not be what you expect. In this case, the Double-Sided  tool, located in the lower part of the dialog page, may help. It changes the line properties of an object from invisible to continuous, enabling all edges to be seen. Otherwise, some surfaces may receive no fill (see Figure 21). For lines without fill, the effect is enabled by default and cannot be disabled. This is also a property of single objects.

1 To reduce both calculation time and data storage, circles are often constructed as regular polygons. If you cut a sphere or cylinder of 10 segments through the middle, you end up with a 20-cornered cut section (10-cornered for a hemisphere).

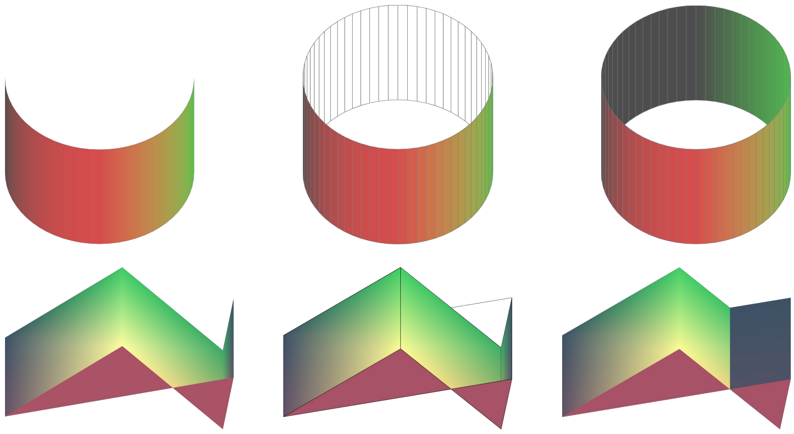


Figure 21: Left: without "Double-Sided"
Middle: with continuous lines but without "Double-Sided"
Right: with "Double-Sided"

Use the buttons in the section *Normals* to modify the normals of a 3D object.

A *Normal* is a straight line that is perpendicular to the surface of an object, in the same way that a vector, starting from an inner point and extending outwards, is at right angles to the surface of the object at the point where it exits the object's surface. Figure 22 shows some normals extending outwards from a sphere made up of 10 segments.

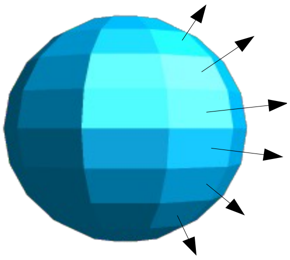


Figure 22: Normals (vectors) of a 3D sphere with 10 segments

Using normals, the display of the object's surface and of its variation in colors, textures, and lighting can be controlled, directly influencing how the surface of the object is rendered.

The first three icons in Figure 14 work as "either-or" switches. Only one of the effects can be active at a given time; an effect can be switched off by clicking on one of the other icons.

Settings belong to individual objects, not scenes; each object can have its own settings. The rest of the icons are normal toggle "on-off" switches. The following effects are available:

Object specific: The object is rendered to produce the best result independent of its shape. Single segments are changed, but their "edges" will hardly be visible.

Flat: The surface of the 3D object will be divided into single polygons, whose edges are clearly recognizable. Every polygon is generated with a uniform color.

Spherical: The enclosing sphere is calculated for the object and then projected onto the object. This form of calculation produces an object with a surface whose segments' edges are more smoothly rounded than can be accomplished with the Object-specific effect (see Figure 23). However, surfaces that meet at a point do not show a realistic lighting effect.

Invert Normals: This results in a reversal of the lighting direction. The inside of the body then becomes the outside. This property is particular to each individual object.

Double sided illumination: The lighting is also computed for the inside of the object. In other cases the lighting value for the outside is simply transferred to the inner side. This property is of interest for open objects. It is a property of the 3D scene and affects all objects in the scene.

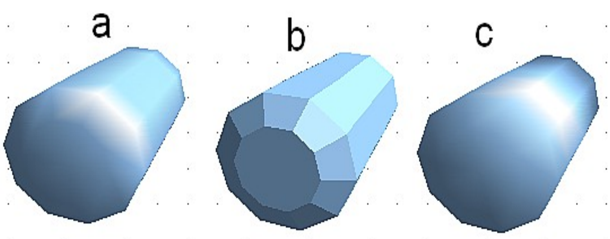
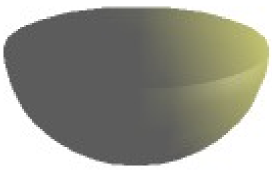
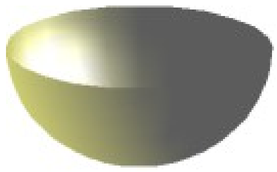
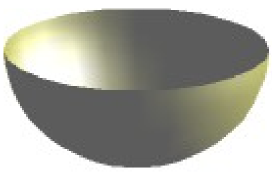
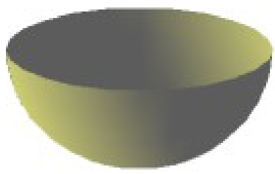


Figure 23: Cylinder with 10 segments
a= Object-specific b=Flat, c=Spherical

The following table shows the linkage between double-sided illumination and inverted normals.

The light source is on the right.	Normals not inverted	Normals inverted
No double sided illumination		
With double sided illumination		

3D Effects - Shading

The Shading page shown in Figure 24 offers functions for shading the object surface, adding shadows, and choosing camera settings.

Shading is a rendering method involving a consideration of lighting ratios. Shading is used to produce curved 3D surfaces. The surfaces are broken down into small triangular segments. Draw offers three methods to produce this effect: *Flat, Phong, and Gouraud*. The setting selected applies to all objects in the 3D scene.

- *Flat* is the fastest and simplest method. For every individual segment, a special color tone is determined based on the lighting ratio and the direction of the segment area. This tone is used for the whole area of the segment. Segmentation is clearly visible.
- *Phong* is the most time-consuming method. With Phong, for each pixel, the associated normal is determined by interpolation, based on the normals of the segment edges. This causes the segment area to appear curved, and the segment intersections are no longer visible.

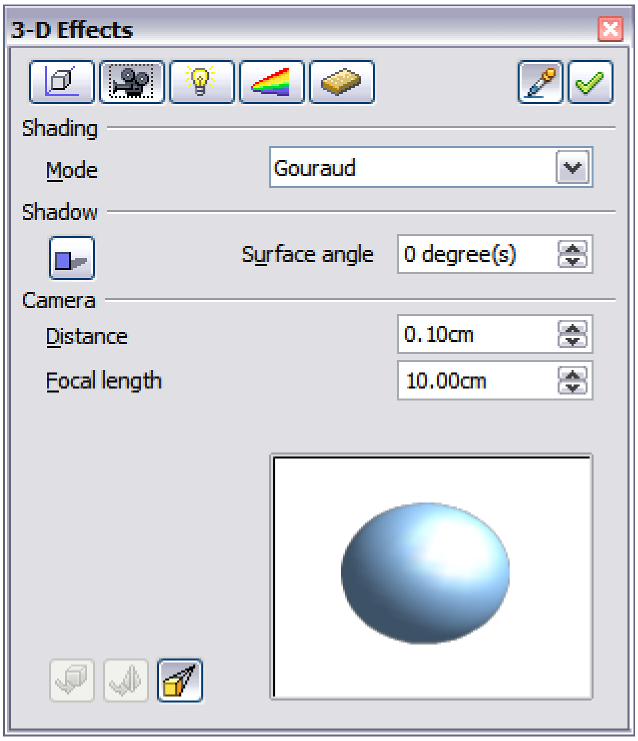


Figure 24: Shading page

- *Gouraud* is a relatively quick method. It determines the color value for the segment corners and calculates the color value for every pixel through linear interpolation. The segment edges are still recognizable, but significantly less so than with the flat method. The Gouraud method considers only light reflection on diffuse, reflecting surfaces.

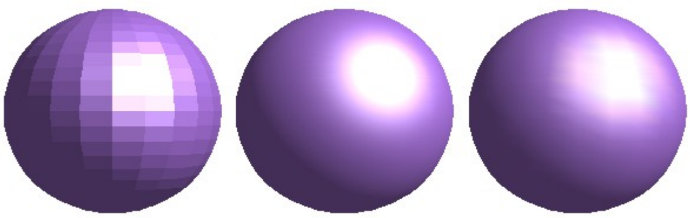
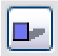


Figure 25: Flat, Phong and Gouraud shading

In Figure 25, the left sphere was rendered with flat shading, the middle with Phong, and the right with Gouraud. The quality of the flat method is clearly inferior to the other two. The difference between Phong and Gouraud is small. With the Gouraud method, the segments can be very faintly seen, and rendered objects have slightly less “shine” than with the Phong method.

All three methods function at the pixel level, and therefore shadowing and mirroring inside the 3D scene (as permitted by ray tracing methods) are not possible..

Using the Shadow button , you can provide a 3D object with a shadow. By changing the *Surface angle*, you can influence the form of the shadow (see Figure 26). The left sphere has a surface angle of 0° (the paper represents a perpendicular surface behind the object), while the right sphere has a surface angle of 45°. At 90°, the paper would be directly under the object.

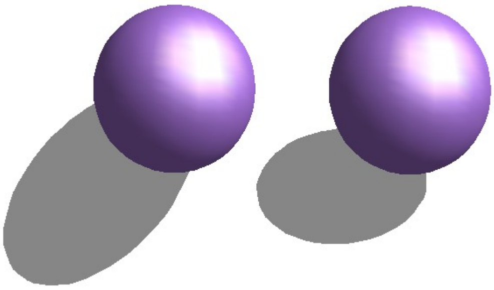


Figure 26: Shadows using different surface angles

The shape and size of the shadow are also influenced by the lighting properties. These can be adjusted on the *Illumination* page of the dialog. Note that multiple light sources for shadows are not supported. Shadow properties can be set for individual objects in a drawing, but where objects form part of a 3D scene, the shadow produced is that of the entire scene.

It is possible to set the shadow property for individual objects using the *Area* item from the menu, brought up with a right-click. The shadow is then shown with the color selected in the shadow dialog, but again, the representation of the lighting of the scene determines the end result of the entire scene. In this way, colored shadows, at varying distances from the object, with different colors and transparency effects, can be created.

In the *Camera* field of the shading dialog, the virtual camera settings can be changed. These settings relate only to the view in central projection and apply to the entire 3D scene. The *Distance* parameter adjusts the spacing between the camera and the scene. The default value for an extruded body is equal to the depth or length of the body. With equal-length edges front and rear, the effect at large distances is quite small. The standard value of the *Focal length* is 10cm (3.937 inches). It has the same significance as a real camera. Larger focal lengths simulate a telephoto lens, smaller ones a wide-angle lens. The effects of changes in camera settings on an object are shown in Figure 27.

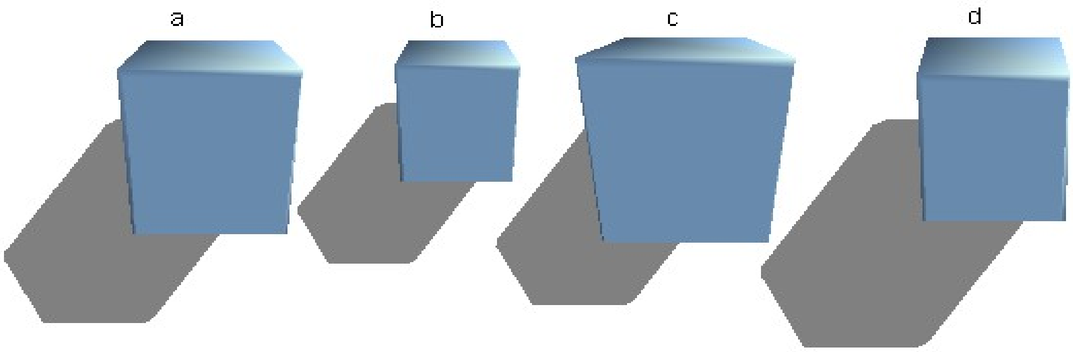


Figure 27: Effect of camera settings

Object a in Figure 27 shows a 3D object with the standard settings. The individual changes to each object are listed in the following table.

	a	b	c	d
Distance:	0.81 cm	3.81 cm	0.81 cm	0.81 cm
Focal length:	10 cm	10 cm	5 cm	15 cm

3D Effects - Illumination

On the Illumination page (see Figure 28), you define how a 3D object is lit. The settings apply to all objects in a scene.

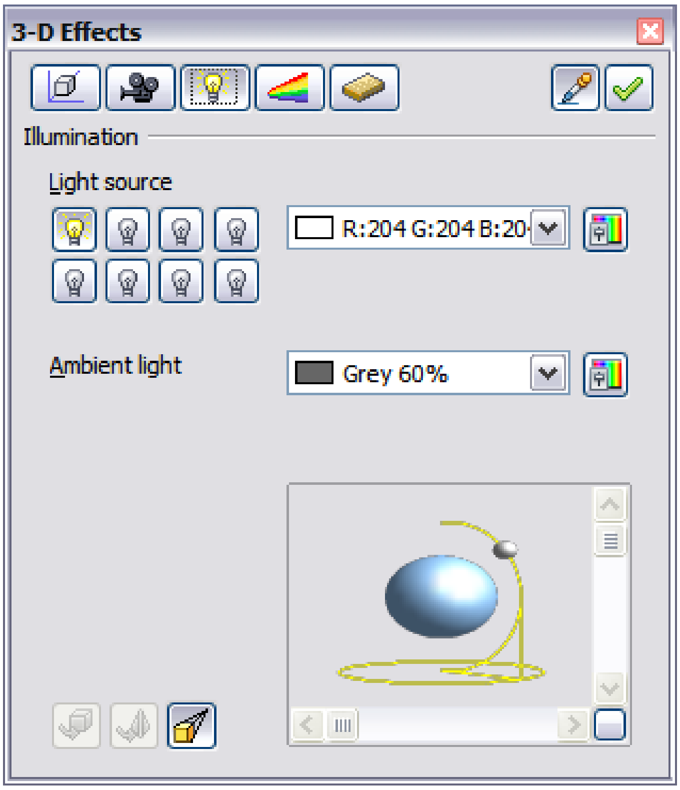


Figure 28: Illumination (lighting) page

You can light a scene with a maximum of eight individual *Light sources* at the same time. For each of these sources, the light color and position relative to the scene can be set. The light sources are represented by eight small light bulbs. When you select this page, the first bulb “lights up”  . At least one light source must be active; otherwise, the rendering and shading functions cannot function correctly.

Each symbol functions like a multi-function press switch. With the first mouse click, the bulb is selected and with the second click, the settings mode for this light source is activated (see Figure 29). A third click deactivates the light source.

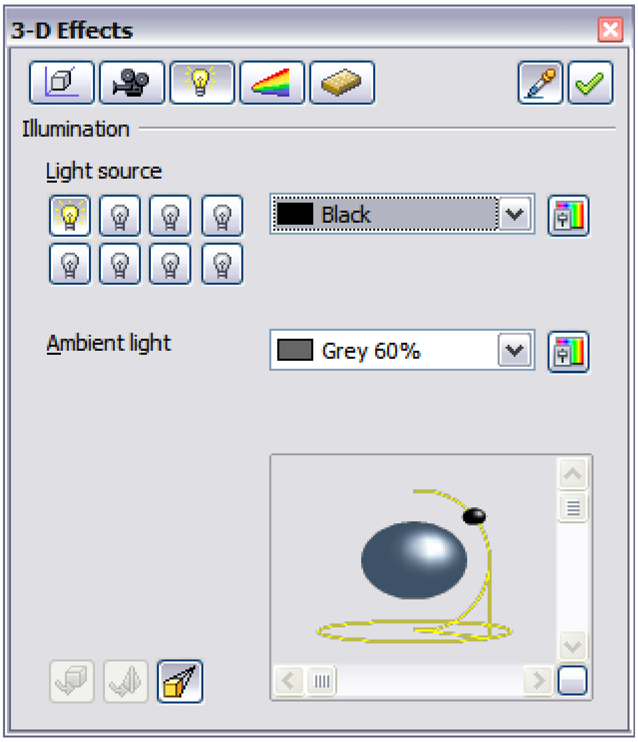


Figure 29: Adjustment of light source

In the selection list next to the symbols, you can choose the color of the active light source. If desired, click the  button to open a color palette dialog, where you can define your own color and also adjust the brightness. For the first light source, it is best to retain the neutral color value (default is white); with several light sources, it is advisable to reduce the brightness.

In the lower-right corner of the menu, the light source location and orientation are depicted. With the vertical slider bar, you can adjust the lighting angle; with the horizontal bar, the light is rotated about the object. Alternatively, you can click the light point and drag it as you please (see Figure 30). Click on the small square in the bottom right (circled) to change the preview image from a sphere to a cube.

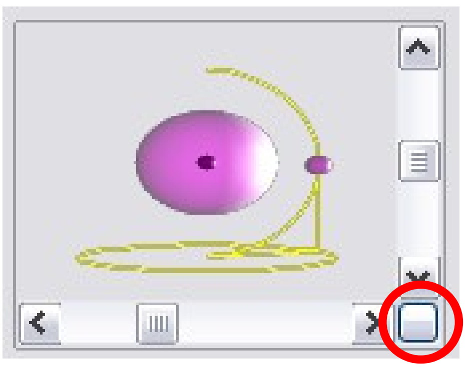


Figure 30: Moving the light source

To apply the changed settings to the selected object, click the **Assign** button .

Using additional light sources can result in some interesting effects.

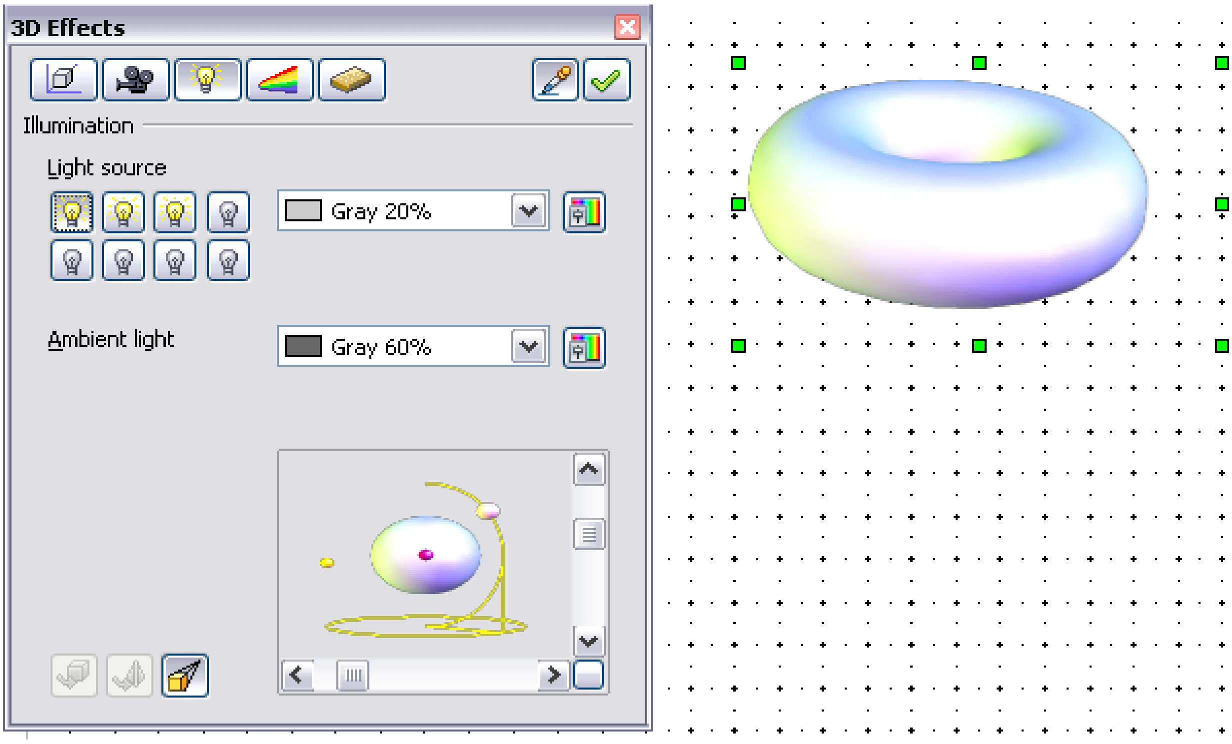


Figure 31: Lighting with three independent light sources

In Figure 31, the ring has the lighting settings from Figure 30 with the standard color white. In addition, it was lit with magenta and, from the left underside, with yellow. The number and position of the light sources are shown in the window at the lower-right. The three light sources in use can be recognized by the “lit” symbols .

To see the effect of a particular lighting effect, you can also temporarily turn it off. With an object selected, clicking a “lit” symbol turns it off . This new setting must then be assigned to the scene. With a further mouse click on the light source, the effect can be switched back on and then re-assigned.

You can also change the settings for the *Ambient lighting*. The selection of properties (lighting color, brightness, and so on) is carried out in the same way as for light sources.

3D Effects - Textures

Textures are raster graphics (bitmaps), that can be used as an object property for the surface of an object. Every object in a 3D scene can have its own texture.

You can set a raster graphic as a texture for a 3D object in the same way as for a 2D object (the Area pane of the Sidebar, the menu **Format > Area > Bitmaps**, or right-click on the object and select Area > Bitmaps) – as is the case for Gradient and Hatching. More details are found in Chapter 4 (Changing Object Attributes).

If the Fill setting in the Area dialog is set to *Color*, the Texture page of the 3D Effects dialog is inactive. Change it to Bitmap to activate the Texture page for 3D objects. If the texture is not tiled or stretched and is smaller than the object, the remaining area will use the color of the Object color property on the Material page.

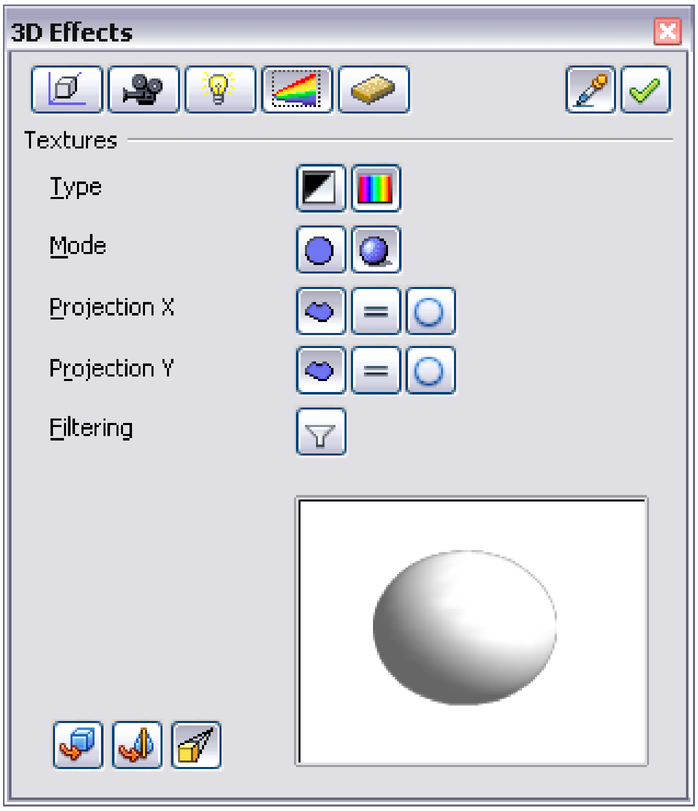


Figure 32: Textures page

In the first row of the page (Figure 32), there are two *Type* switches that allow you to choose between black-and-white or color for the texture. In Figure 33, the texture has been applied to spheres b and c using the Gallery. Sphere b has the full color pattern, and the pattern in sphere c has the texture converted to black and white using the 3D Effects dialog.

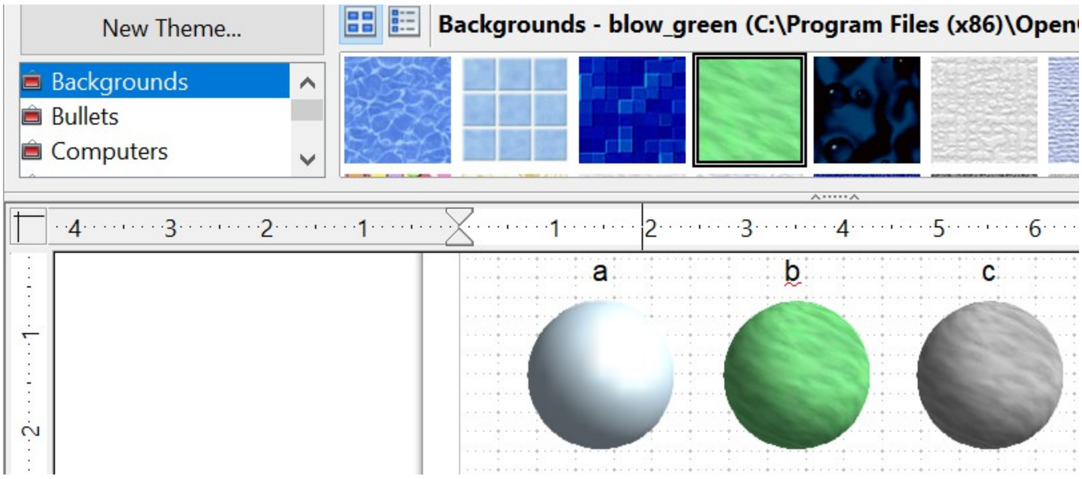


Figure 33: a - standard color setting, b - texture (color), c - texture (black and white)

With the two switches in the row *Mode*, you can control whether the texture of the selected objects is rendered with light and shadows (Switch 2) or not (Switch 1); see Figure 34. Appropriate lighting and shading adjustments allow the graphic object to be rendered more realistically.

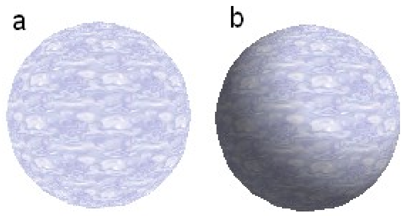
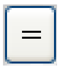
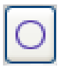


Figure 34: Texture without (a) and with (b) lighting and shadow effects

Projection X / Y

With one of these three buttons, you can determine how the texture for this coordinate axis should be projected onto the object. The default setting, *Object specific,* usually gives the best results. Examples of the use of each button are shown below.

	Object specific	The texture is automatically adjusted for the best fit with the object's form and size.
	Parallel	The texture is projected parallel to the object's axis. It is mirrored on the object's rear side.
	Circular	The axis of the texture pattern is wrapped around the object.

For a rotation body (Figure 35), the turning axis is the Z-axis and the wrapping is the X-direction; for an extrusion body (Figure 36), the extrusion direction is the Z-axis and the extruded surface is the X-direction. Different texture positions result depending on how the object was produced.

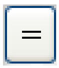
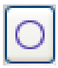
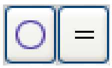
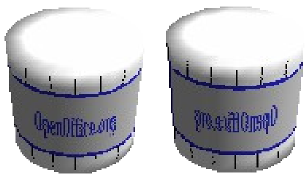
 Area fill without tiling, with adjustments	Projection X 	Projection X 	Projection X 
Projection Y 			
Projection Y  (Difference is small)		 Front and rear side	

Figure 35: Cylinder as a rotation body

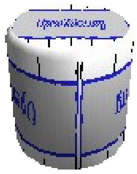
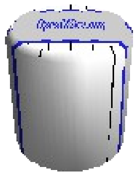
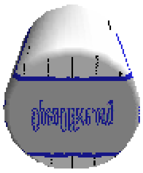
 Area fill without tiling, with adjustments	Projection X	Projection X	Projection X
Projection Y			
Projection Y (Difference is small)		  Upper and under sides	

Figure 36: Cylinder as an extrusion object

The *Filter* button switches on and off a soft-focus filter. It can often remove slight faults and errors in the texture.

3D Effects - Material

On this page (Figure 37), you can assign the appearance of different materials to the 3D object.

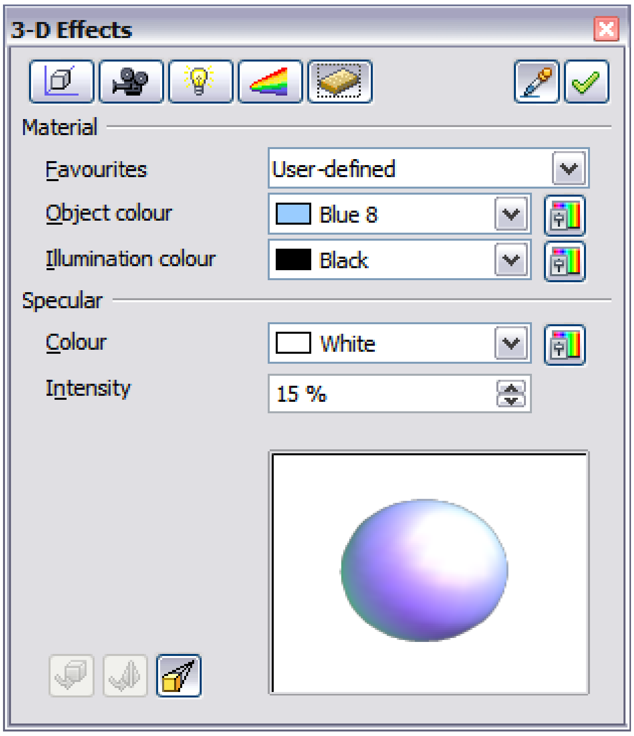


Figure 37: Material menu

In the selection list under Favorites, are the most commonly used materials (see Figure 38).

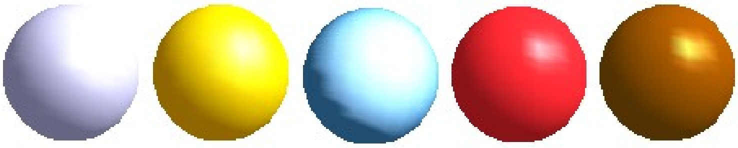


Figure 38: The favorites (from left to right): Metal, Gold, Chrome, Plastic and Wood

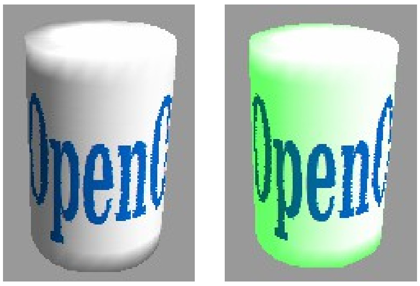
With the individual *color parameters*, additional effects can be produced. The meaning of these parameters corresponds to those on the Illumination dialog page.

Materials and textures can be combined. Settings only simulate materials, so achieving the desired result can sometimes require trial and error.

Tip	Do not use a brightness value that is too high for individual colors. These are additive, and it's easy to end up with a totally white “colored” area.

The *Illumination color* brightens parts of the object that are in shadow, making the body appear more illuminated.

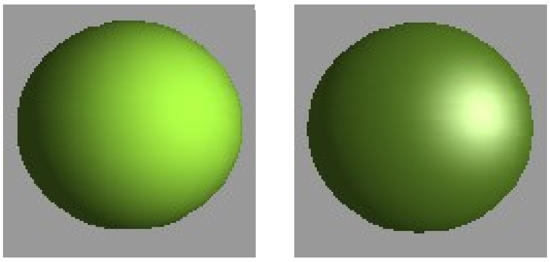
When textures are used as well, the Illumination color is combined with the white color part of the texture. On the left, the object has a black Illumination color, on the right, bright green.



The *Specular color* simulates the reflecting capacity of the surface. The position of the illuminated point is determined by the setting of the first light source.

Left: Set the Specular color the same as the object color and the illumination point intensity to a low value, to create the impression of a matte body.

Right: Set the Illumination point color the same as the light source color, in order to give a shiny appearance to the object's surface.



Metallic surfaces and glass are poorly simulated because their appearance is produced by reflection. Such a simulation cannot be performed by OpenOffice.

3D Settings for 3D Shapes

3D shapes are handled quite differently from objects in 3D scenes. The appearance of a shape object is changed using the *3D-Settings* toolbar (see Figure 39). The *3D*

Effects dialog described above should not be used for shape objects, as it will not give the correct formatting results.

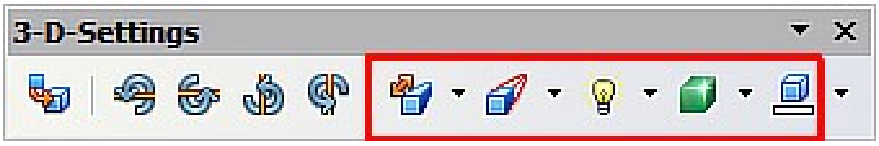
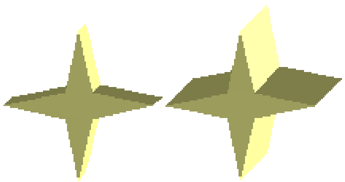


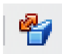
Figure 39: 3D-Settings toolbar, indicating icons for formatting 3D shapes

If you have used it in error, you can remove the incorrect formatting with **Format > Default Formatting**.

Using the buttons on this toolbar, you can adjust the extrusion depth, perspective, lighting, material properties, and the extrusion color. There are tear-off bars that you open by clicking the small black triangle. The purpose of each button is described in its tooltip. You do not need to assign the result of any of the functions to the object as you do with the 3D Effects dialog used with 3D scenes; every action is immediately applied, and you can see its effect on the object in the main Draw window.

Here are some examples of the formatting of 3D shapes:

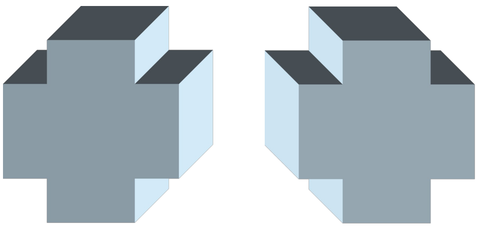


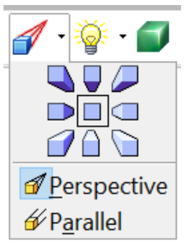
Depth 

The tool has six preset values from zero to infinity, plus a custom option to set any value.

left: 0.3cm

right: 1cm

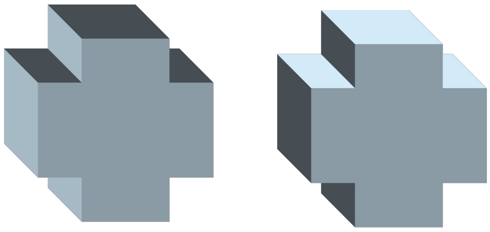


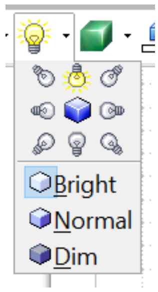


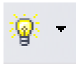
Direction 

The tool allows you to change the direction from which the object is viewed, and between parallel and perspective views.

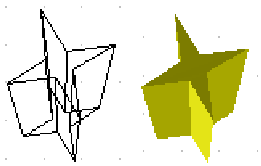
The examples show a cross after choosing the lower-left or lower-right icon from the nine direction options in the tool.

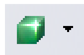




Illumination 

The tool provides eight illumination directions and three light levels. The examples show an object illuminated from the lower-left and the top.

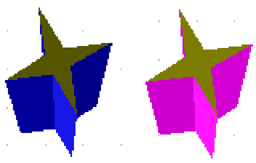


Surface 

Only four built-in variations are possible. At present, only *Wireframe* and *Matt* are correctly rendered.

left: *Wireframe*
right: *Matt*

You can also choose a gradient, hatching, or bitmap for the surface; these are only applied to the extruded surface—the sides remain in the color of the object.



3D Color 

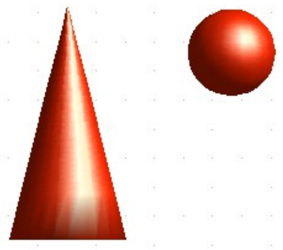
Here, you can choose the color of the sides of the extruded surface.

The symbol shows the color of the most recently chosen shape object.

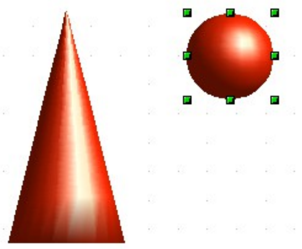
Combining Objects in 3D Scenes

3D objects produced by extrusion or rotation are shown in the status bar as a *3D scene*. You can group several of these objects together. Other object types cannot be grouped in this way. Management of the group is carried out in the same way as described elsewhere in this guide (**Modify > Enter Group** or **Modify > Exit Group**). Also see “Grouping Objects” in Chapter 5 (Combining Multiple Objects).

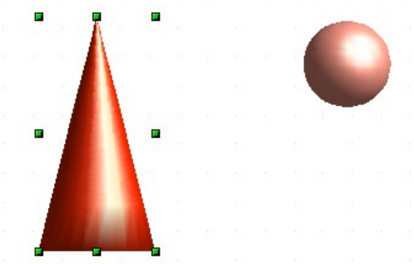
As an example, we will produce a game piece.



First, produce both objects independently of each other. The subsequent combination is made easier if you use parallel projection and rotate the objects into an upright, straight position.

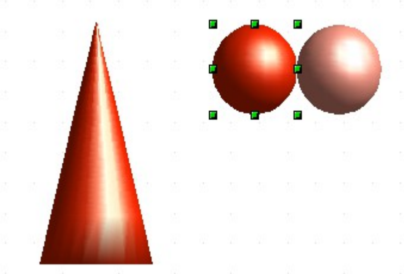


Click on the sphere and use **Edit > Copy** to take over the scene and put the sphere on the clipboard. If you are sure you no longer need the original, you can use **Edit > Cut**. In any case, move the sphere a little to the side.

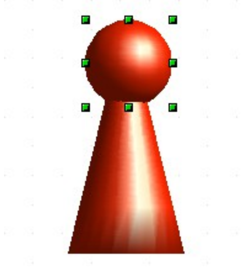


Click now on the cone. The status bar will show *3D Scene selected*. Right-click and enter the group. You will see that, as usual, elements that do not belong to the group are less bright.

Use **Edit > Paste**. Now the objects (not the whole scene) from the clipboard are pasted into this scene.

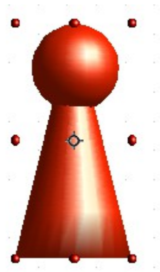


The original position of the sphere now contains a new sphere that belongs to the scene. This new sphere can be dragged over to the cone. Exit the group after moving the sphere.

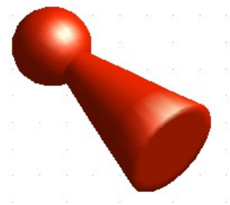


Notice that combining 3D objects results in them being more or less “fused” together – there is no stacking or layering as with 2D objects.

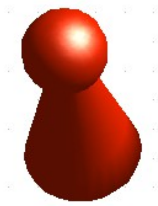
Adjust the position of the objects as you wish. You cannot arrange objects in front of or behind others, as with 2D objects, but can only move them parallel to the drawing plane.



Enter the group again and adjust the objects. The status bar indicates which object is marked. Use the *Tab* key to change from one object to the next in the group if it is not possible to do so with the mouse. Exit the group.



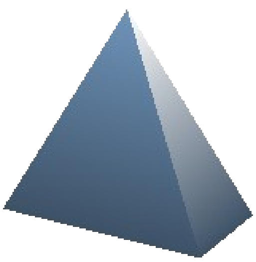
Now you can rotate the entire 3D scene and view your game piece from all sides.



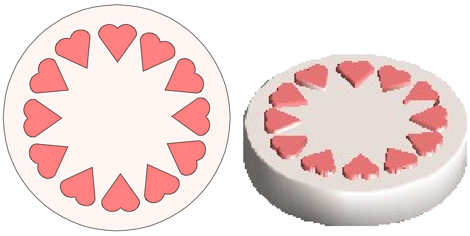
Examples for Your Own Experiments

All examples use objects in 3D scenes.

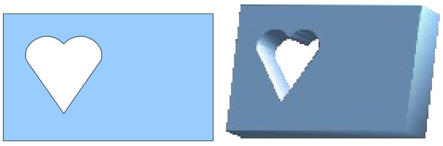
Extruding a 2D object with text generates letters as separate objects; they have a larger extrusion depth than the background.



3D objects of the *Shape* type can be rendered as wireframe models. This effect can be produced in other 3D objects by setting the Area fill to *None* and the line style to *Continuous*.

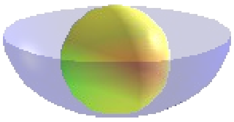


If you select several 2D objects—without grouping them—and extrude the selection, they are transformed, according to their stacking order, with different extrusion depths. The previously topmost object will be uppermost on the extruded object.



For 2D objects, use **Modify > Shapes > Merge/Subtract/Intersect** to produce complex objects. The resulting figures can be extruded, rotated, and so on.

Rotation of lines produces concave bodies. Use a bright line color. With a high number of vertical segments, the transition to the ground is relatively sharp-edged. Remember to switch on two-sided illumination.



The transparency of the platter is adjusted in the surface properties of the body. The best effect is obtained only when the transparent body is combined with other objects.

The fruit's color shadow is produced using multiple light sources; they are not “real” and do not rotate or change as the scene rotates.